MUHAMMED AFNAN S

Unity Developer | Unity XR Developer | Game Developer

Dubai, UAE • Phone: +971 58 143 5669 • Email: muhammedafnans6913@gmail.com

Portfolio: https://afnan-portfolio-eight.vercel.app/

SUMMARY

Unity Developer with 2+ years of professional experience developing mobile games, multiplayer systems, and XR applications using Unity and C#. Experienced in 2D, 2.5D, and 3D game development, VR, AR, Mixed Reality (MR), WebSocket-based multiplayer, Netcode, Photon, gameplay systems, UI, physics, AI/bot systems, API integration, and mobile optimization. Experienced in developing applications for Android, iOS, PC, and WebGL, with hands-on experience in Git/GitHub, debugging, profiling, and performance optimization.

CORE COMPETENCIES

Unity Development, C# Programming, Game Development, XR Development, VR, AR, MR, Multiplayer Development, Netcode, WebSocket, Photon, Gameplay Programming, AI/Bot Systems, Physics, UI Development, Mobile Game Development, Android, iOS, API Integration, Performance Optimization, Debugging, Git, GitHub

WORK EXPERIENCE

XR Developer — Infusory Future Tech Labs Pvt Ltd, Kerala, India *March 2026 – June 2026*

- Developed XR applications using Unity and C# for Mixed Reality (MR) learning experiences.
- Designed and implemented interactive XR activities, spatial interactions, object tracking, and user input systems.
- Developed reusable Unity components and interactive systems for XR applications.
- Worked with VR/AR/MR workflows and real-time interactive experiences.
- Integrated multiplayer/networking functionality using Netcode and networked gameplay systems.
- Optimized Unity XR applications for smooth performance and real-time interaction.
- Collaborated with cross-functional teams to develop and deliver scalable XR applications.

Unity Game Developer — Bagheera Carrom OPC Pvt Ltd *July 2025 – January 2026*

- Developed and maintained multiplayer mobile games using Unity and C#.
- Developed Bagheera Ludo Multiplayer across the complete game development lifecycle.
- Implemented gameplay mechanics including dice systems, token movement, turn management, validation, and win conditions.
- Integrated WebSocket real-time multiplayer communication for matchmaking, gameplay synchronization, and reconnection.
- Developed AI/bot gameplay systems for offline and fallback gameplay.
- Integrated REST API services, in-app purchases, virtual currency, and game data systems.
- Optimized gameplay, touch controls, physics, UI, and performance for Android and iOS.
- Used Git and GitHub for version control, collaboration, debugging, and project management.

Associate Game Developer — Emergio Games Pvt Ltd *July 2023 – May 2025*

- Developed Unity-based 2D, 2.5D, 3D, and multiplayer games using C#.
- Implemented gameplay mechanics, UI systems, animations, physics, and interactive game systems.
- Developed and debugged gameplay features and resolved UI, performance, and gameplay issues.
- Built and tested Android mobile game builds for stability and performance.
- Optimized gameplay systems and Unity projects for improved runtime performance.
- Collaborated with designers and developers in an Agile development environment.
- Used Unity, C#, Visual Studio, Git, and GitHub for daily development.

TECHNICAL SKILLS

- Game Engine: Unity, Unity 2D, Unity 2.5D, Unity 3D
- Programming: C#, C++ (Basic)
- XR: Unity XR Development, Mixed Reality (MR), Virtual Reality (VR), Augmented Reality (AR), Interactive XR Applications, Spatial Interaction, Object Interaction, Meta Quest
- Multiplayer & Networking: Netcode, Photon, WebSocket, Online Multiplayer
- Game Development: Gameplay Programming, Game Mechanics, AI/Bot Logic, Physics, Animation, UI Systems
- Mobile Development: Android, iOS, Mobile Optimization, Performance Optimization
- Platforms: Android, iOS, PC, WebGL
- Backend & Integration: REST API, API Integration, WebSocket Communication, In-App Purchases, Virtual Currency
- Tools: Git, GitHub, Visual Studio, Unity Profiler

PROJECTS

Bagheera Ludo Multiplayer (Unity | Online Multiplayer)

- Developed a real-time online multiplayer Ludo game using Unity.
- Implemented turn-based gameplay mechanics, dice logic, token movement, and win conditions.
- Integrated WebSocket communication for matchmaking and gameplay synchronization.
- Designed bot/AI logic for offline and fallback gameplay.
- Integrated in-app purchases and virtual currency system.
- Optimized performance for Android and iOS platforms.
- App Store Link: https://apps.apple.com/in/app/bagheera-ludo/id6754906087

Bagheera Carrom (Unity | Board Game)

- Implemented realistic striker physics and collision handling using Unity physics engine.
- Improved shot accuracy and control through fine-tuned input and force mechanics.
- Optimized collision detection and rigidbody interactions for consistent gameplay.
- Fixed critical gameplay bugs related to coin movement, pocket detection, and scoring.
- Enhanced UI responsiveness and gameplay feedback for better user experience.
- Worked on turn-based logic and rule enforcement for authentic carrom gameplay.
- Reduced frame drops by optimizing physics calculations and object updates.
- Assisted in polishing game feel through animation and sound integration.
- App Store Link: https://apps.apple.com/in/app/bagheera-carrom/id6746533129

American Roulette (Unity | Casino Game)

- Implemented realistic wheel physics and ball spin behavior to simulate real casino experience.
- Developed probability-based outcome system ensuring fairness and randomness.
- Designed modular betting architecture supporting multiple bet types (inside, outside bets).
- Integrated dynamic UI updates for real-time betting feedback and results display.
- Implemented chip management system with balance tracking and bet validation.
- Optimized game performance for smooth animations on low-end mobile devices.
- Handled edge cases such as invalid bets, insufficient balance, and game state resets.
- Improved user engagement through responsive UI/UX and animation transitions.

XR Educational Applications / Interactive Activities (Unity | MR | Netcode | Meta Quest)

- Developed reusable Mixed Reality (MR) educational activities in Unity for interactive student learning experiences.
- Created object-based interactions allowing students to interact with and explore virtual learning objects.
- Built activities on an existing multiplayer architecture using Unity Netcode for shared student-teacher experiences.
- Supported multiplayer XR sessions where students and teachers can interact within the same virtual room.
- Tested and validated XR activities on Meta Quest devices.

Additional Projects

- Two Endless Runner games
- Monopoly-style board game prototype
- Two small free-world game demos

EDUCATION

BSc Mathematics — Kerala University (2020 – 2023)

LANGUAGES

English, Malayalam, Tamil, Hindi